

ASSIGNMENT 1 – INTRODUCTION TO WEB

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Question 1: Explain the difference between frontend, backend, and full-stack development with suitable real-world examples.

Answer:

Frontend development is the part of a website that users can see and use. It includes buttons, images, menus, and pages. HTML, CSS, and JavaScript are mainly used for frontend.

Backend development works behind the scenes. It manages data, user accounts, databases, and server operations. Languages like Java, Python, PHP, and Node.js are used.

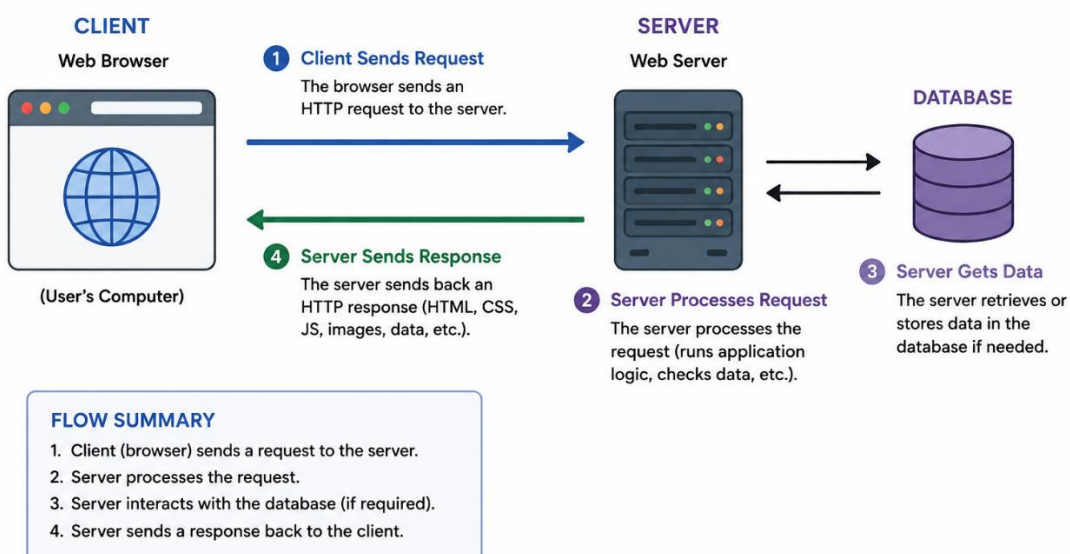
A Full Stack Developer works on both frontend and backend.

Example: In Amazon:

- Frontend: Product page and shopping cart.
- Backend: Stores product details, user information, and orders.
- Full Stack: Develops both frontend and backend.

Question 2: Create a simple diagram showing how the client-server model works in web architecture.

Client-Server Model in Web Architecture



Question 3: Describe how a browser requests and displays a web page from a web server.

Answer:

1. The user enters a website address in the browser.
2. The browser sends a request to the web server.
3. The server processes the request.
4. The server sends HTML, CSS, and JavaScript files.
5. The browser displays the web page on the screen.

Question 4: Identify and list the tools required to set up a web development environment. Explain the purpose of each.

Answer:

Tool	Purpose
VS Code	Writing and editing code
Web Browser	Testing websites
Git	Tracking code changes
GitHub	Storing projects online
Node.js	Running JavaScript outside the browser
Live Server Extension	Automatically refreshes the webpage

Question 5: Explain what a web server is and give examples of commonly used servers.

Answer:

A web server is software that stores website files and sends them to users when they visit a website.

Examples:

- Apache
- Nginx
- Microsoft IIS
- Node.js Server

Question 6: Define the roles of a frontend developer, backend developer, and database administrator in a project.

Answer:

Frontend Developer

- Designs the user interface.
- Makes websites responsive.
- Uses HTML, CSS, and JavaScript.

Backend Developer

- Creates server-side logic.
- Connects the website with the database.
- Manages user authentication.

Database Administrator (DBA)

- Creates and manages databases.
- Keeps data secure.
- Performs backup and recovery.

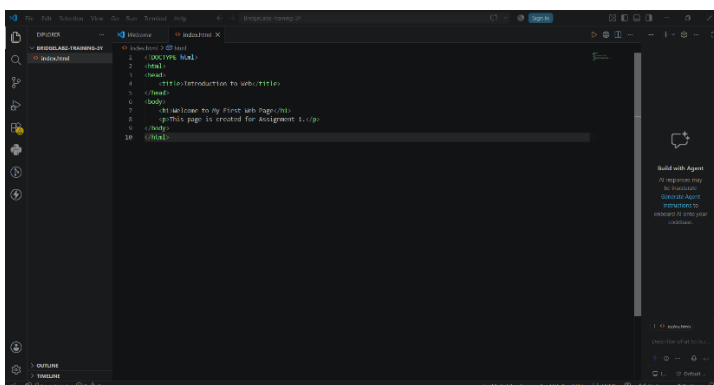
Question 7: Install VS Code and configure it for HTML, CSS, and JavaScript development. Take a screenshot of the setup.

Answer:

I installed Visual Studio Code and configured it for web development.

Installed Extensions:

- Live Server
- Prettier
- HTML CSS Support



Question 8: Explain the difference between static and dynamic websites. Provide an example of each.

Answer:

Static Website	Dynamic Website
Fixed content	Content changes automatically
Faster	Interactive
No database	Uses database

Example of Static Website: Portfolio website

Example of Dynamic Website: Facebook, Amazon

Question 9: Research and list five web browsers. Explain how rendering engines differ between them.

Answer:

Browser	Rendering Engine
Google Chrome	Blink
Microsoft Edge	Blink
Mozilla Firefox	Gecko
Apple Safari	WebKit
Opera	Blink

Rendering engines read HTML and CSS and display web pages. Different browsers use different rendering engines, so websites may look slightly different.

Question 10: Draw a labeled diagram showing the basic web architecture flow — client, server, database, and APIs.

Answer:

BASIC WEB ARCHITECTURE FLOW

